

AI Meal Planner & Smart Grocery Assistant

GitHub Repository:

<https://github.com/rileyruiggiero-boop/AI-Meal-Planner>

1. Introduction

Meal planning is a common problem for many people. It can be time-consuming and difficult, especially when trying to balance health goals, budget constraints, and dietary restrictions. Many existing tools either provide generic meal suggestions or require manual searching through recipes, which does not efficiently solve the problem.

This project, **AI Meal Planner & Smart Grocery Assistant**, was created to address this issue by combining real-world recipe data with generative AI. The goal of the project is to allow users to quickly generate personalized meal plans based on their preferences, while also receiving intelligent recommendations and a structured grocery list.

The application provides a working minimum viable product (MVP) that integrates external APIs and generative AI into a single, functional system that solves a real-world problem.

2. Problem Statement

Planning meals manually presents several challenges:

- It takes time to find recipes that match goals (e.g., high protein or healthy)
- Users must manually check ingredients and instructions
- Budget considerations are often ignored
- People with dietary restrictions have limited options
- Grocery list creation is repetitive and inefficient

The objective of this project is to create a system that automates these tasks and enhances them using generative AI.

3. Solution Overview

The solution is a web-based application built using Streamlit that allows users to generate personalized meal plans.

Users can input:

- Meal goal (e.g., high protein, healthy)
- Daily calorie target
- Grocery budget
- Number of days for the plan
- Dietary restrictions (e.g., vegetarian, vegan)
- Ingredients to avoid

The system then:

1. Uses the Spoonacular API to retrieve real recipe data
2. Extracts ingredients, instructions, and nutritional information
3. Processes the data into a structured format
4. Sends the data to the OpenAI API
5. Generates a detailed AI-powered explanation, scoring, and recommendations
6. Produces a downloadable grocery list (TXT and CSV)

This creates a complete end-to-end workflow from user input to final output.

4. Technologies Used

This project uses the following tools and technologies:

- **Python** – main programming language
 - **Streamlit** – web application framework
 - **Spoonacular API** – provides recipe and nutrition data
 - **OpenAI API** – generates explanations, scoring, and recommendations
 - **Requests library** – handles API calls
 - **python-dotenv** – manages environment variables securely
 - **Git & GitHub** – version control and project hosting
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5. Generative AI Integration

Generative AI is a core component of this project.

The OpenAI API is used to:

- Generate a structured meal plan summary
- Explain why the plan fits the user's goal
- Evaluate whether the plan fits the user's budget
- Create a categorized grocery list
- Provide improvement suggestions
- Assign scores for:
 - Nutrition
 - Budget
 - Convenience
 - Overall quality

This demonstrates clear use of generative AI beyond simple text generation. The AI processes structured data from an external API and transforms it into meaningful, user-friendly insights.

6. System Workflow

The system operates in the following steps:

1. User inputs preferences into the Streamlit interface
2. The app sends a request to the Spoonacular API
3. The API returns a weekly meal plan
4. The app retrieves detailed recipe information for each meal
5. Ingredients and instructions are displayed to the user
6. A combined ingredient list is generated and cleaned
7. The data is sent to the OpenAI API
8. The AI generates a structured explanation and recommendations
9. The user can download a grocery list

This workflow demonstrates a complete and functional MVP.

7. Features

Key features of the application include:

- Multi-day meal planning
- Dietary restriction filtering

- Ingredient exclusion
- Budget input and evaluation
- Real recipe ingredients and instructions
- AI-generated explanations and recommendations
- AI scoring system
- Downloadable grocery lists (TXT and CSV)
- Clean and interactive user interface

These features contribute to a strong and practical application.

8. Challenges and Limitations

Several challenges were encountered during development:

API Limitations

The Spoonacular API sometimes restricts access or limits the number of requests, which required careful handling of API calls.

Data Inconsistency

Some recipes did not include full instructions or consistent ingredient formats, requiring fallback handling.

Formatting Issues

Combining data from APIs and ensuring clean output required multiple iterations.

Debugging

Errors such as indentation issues, API errors, and environment setup problems were common and required troubleshooting.

9. Results

The final application successfully meets the project objectives:

- The app runs correctly using Streamlit
- It integrates two APIs (Spoonacular and OpenAI)

- It produces meaningful outputs based on user inputs
- It provides a complete meal planning workflow
- It demonstrates clear use of generative AI

The application is functional, practical, and easy to use.

10. Conclusion

The AI Meal Planner & Smart Grocery Assistant demonstrates how generative AI can be combined with real-world data to solve everyday problems.

By integrating APIs, user input, and AI-generated output, the project provides a complete and useful application. It highlights the potential of generative AI in building tools that improve efficiency, decision-making, and user experience.

Overall, the project meets all requirements for a working MVP and showcases practical implementation of generative AI concepts.